

Radar, terrain and VD after the A380 / A320 manuals

VD cut along the flight plan, FCOM scale, terrain by true height, real-display look

Branch	feature/nd-manuals-pass
Aircraft	A32NX and A380X
Commits	15 commits, tip ddca9fe22 (PR 9)
Base	feature/nd-wxr-mode-label (chain top)
Status	In develop. Flown. PR draft 09.

Specification

Rule	Source	Implementation
VD cut along the active F-PLN in managed modes, along the track otherwise; width 2 x RNP	A380 FCOM DSC-31-20-40-10	L :A380X_VD_CUT_* LVars from EfiTawsBridge; gauge draws along it
Scale origin shows the mock-up and most of the trajectory	FCOM DSC-31-20-40-10	aircraft / path at 3/4 of the window
Terrain height under the mock-up geometrically correct, not baro dependent	A380 FCTM SI-70	range from the ADR baro altitude
No magenta on the VD; no VD weather without TERR	FCOM DSC-31-20-40-10	VD weather layer
Radar scans on the ground	A320 FCOM DSC-34-SURV-30-30	ground inhibit removed
MAP mode: black water, green ground, amber mountains	A320 FCOM WXR panel	terrain-view ground map
30 s buffer fill	A380 FCOM WXR operational recommendations	one slow sweep
Real antenna sweep (A320 RDR-4B 180 deg, A380 RDR-4000 160 deg)	vendor documents	cone
No weather above 320 NM	A380 FCOM DSC-31-20-50	range limit

The `ndwxr` module was split into nine files by concern (quality pass), zero warnings for both aircraft.

PR description (from the PR draft)

From `pr-drafts/09-a380x-vd-cut.md`, section "Summary of Changes" onwards.

Summary of Changes

- **Vertical cut (FCOM DSC-31-20-40-10):** `EfisTawsBridge` publishes the cut as `L:A380X_VD_CUT_*` LVars: along the active flight plan in the managed lateral modes (a vertex per path vector, the midpoint of each turn, a 160 NM straight extension, a 200 NM window from just behind the aircraft, twice a second), along the track (IR true track) in the selected modes, as the FCOM's "VIEW ALONG ACFT TRK". The native gauge draws the terrain profile piece by piece along it (falls back to the track when the aircraft is more than 5 NM off the plan), with a width of 2 x RNP (1 NM terminal, 2 NM en route above 18 000 ft) as the highest terrain across five columns, and the grey area from the FMS's next track change > 3 deg. The weather on the VD follows the same cut with a zero width (beam-width sampling).
- **Scale (FCOM "the FMS defines the origin of the vertical scale in order to display the aircraft mock-up and most of the aircraft trajectory"):** with a trajectory the highest of aircraft / path sits at three quarters of the 4 deg window, the top is capped at 70 000 ft and the bottom may go below sea level, so the ground stays on the plot in cruise as on the real aircraft (photo at FL400). Without a trajectory the aircraft stays in the middle.
- **Terrain placement (FCOM / FCTM SI-70 "the height of the aircraft above the terrain displayed on the VD is geometrically correct and does not vary with the altimeter setting"):** the terrain view's range is set from the ADR baro altitude, so the gap under the mock-up is the true height whatever the baro setting. Water is drawn from a water-mask view in the same teal as the ND's terrain display (the FCOM says blue; the ND's colour is what the crew sees next to it). Magenta is not drawn on the VD (FCOM), and the VD weather is removed while TERR is unavailable (FCOM: "the VD does not display the weather when the TERR function is not available").
- **Weather picture (both aircraft, from the manuals pass):** the radar scans on the ground too (A320 FCOM DSC-34-SURV-30-30), MAP mode is a ground map from the terrain view, the picture fills over 30 s after the radar starts transmitting (A380 FCOM operational recommendations), the ARC picture keeps the ND's bottom corners free as on the real display, water on the terrain display is the real display's teal, and the TERR peaks box figures come from SimBridge over SimConnect (PR 4 draft). The A380's AUTO mode off-path display is NOT modelled: it was built from a third MapView in the engine's top-view radar mode and removed again, because the engine runs one radar per aircraft and that view changed what the beam views delivered (rain at the aircraft's level over the whole range instead of the cells; the A32NX next to it was clean). Both aircraft show the same picture.

Cockpit API Changes

New LVars, documented in `fbw-a380x/docs/a380-simvars.md`: `L:A380X_VD_CUT_MODE` (0 track / 1 plan), `L:A380X_VD_CUT_COUNT`, `L:A380X_VD_CUT_{i}_LAT` / `_LON` ($i < 32$, degrees), `L:A380X_VD_CUT_TRACK_CHANGE_NM`. Reads the IR true track words.

Screenshots (if necessary)

TO ADD: the VD through a turn in NAV (profile follows the route, grey area at the turn), the same in HDG (along the track), cruise at 160 NM (ground visible, aircraft in the upper quarter), an approach with terrain (gap under the mock-up with QNH vs STD).

References

- A380 FCOM DSC-31-20-40-10 (VD): vertical cut along the active flight plan / current track, cut width 2 x RNP, vertical scale origin, terrain by true height, water blue, magenta not displayed, weather needs TERR (PDF pages 2365-2394 of the KAL FCOM).
- A380 FCTM SI-70 Safety altitude and terrain / Vertical trajectory / Terrain (VD height geometrically correct; MORA/MSA along the F-PLN in NAV, along the track in HDG/TRACK; grey area = next turning point).
- Real A380 photo at FL400 (ground on the VD, aircraft at ~75 % of the height).

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. NAV mode with a turn ahead, VD on: the terrain profile follows the route through the turn (compare with the ND terrain along the green line); a grey area starts at the turn. Switch to HDG: the profile is along the track; back to NAV: along the plan.
2. Cruise at 160 NM: the ground and the terrain profile are visible, the aircraft symbol in the upper quarter of the VD.
3. Approach with terrain nearby, QNH then STD: the gap under the aircraft symbol does not change.
4. Water on the cut: blue from sea level down. TERR SYS OFF: the VD weather goes with the terrain.